EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Aim

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Gym Membership application. The application is used for managing gym membership registrations, and it served as the practical environment for exploring the security and governance features described in this record. The Zoho Creator account was on an expired trial, so some features required an upgrade; this is documented in the relevant sections.

Security and Governance Architecture

The security and governance structure of the Gym Membership application can be understood in three parts. This is a conceptual description and not a view of the Zoho Creator interface.

- Users and access: users such as the CEO, users with Read / Write access and portal users access the application.
- The application: it contains the forms and data (Gym Membership), security and permissions (user permissions, roles and data sharing) and customisation items (widgets, templates and variables).
- Governance features: user permissions, roles, data sharing and portal user permissions manage and control access to the application.

The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing, and efficient management.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application provided two predefined permissions:

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Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the application, the default role CEO is displayed, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

The Add Role option was used to prepare a custom role with the following details:

Field	Value
Role Name	Gym Manager
Reports To	CEO
Description	Manages gym memberships
Share Data with Peers	Not selected
Status	Not saved – an upgrade was required

The Gym Manager role configuration was prepared with the required name, reporting hierarchy and description. However, saving the custom role required an upgrade because the current Zoho Creator trial had expired. The Gym Manager role was therefore not created, and the CEO role remains the only role in the application.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. A data sharing rule was configured and successfully saved with the following details:

Parameter	Value
Rule Name	Gym Membership Data Sharing
Form	Gym Membership
Shared From	CEO (role)
Shared To	Athithya R (user)
Access Level	Read/Write

The saved rule appears in the Data Sharing list with the columns Name, Form, Shared From, Shared To and Level of access. With Read/Write access, the user Athithya R can both view and modify the shared records of the Gym Membership form that belong to the CEO role. The level of access determines how much a recipient can do with the shared records, so that visibility is extended in a controlled manner.

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Security Features and Governance Controls

The security and governance controls of Zoho Creator can be grouped as User Access Control (user permissions, Read and Write permission, role-based access and portal user permissions), Data Governance (data sharing, read / write access, record visibility, role hierarchy and controlled data access) and Application Governance (localization, contextual messages, record templates, schema builder, application IDE, widgets and variables). The grouping is conceptual and does not show that every listed feature was configured.

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and roles were examined, and a data sharing rule was configured. Portal user permissions were restricted by the account, and the custom role could not be saved. The other items in the grouping, such as localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was accessed for examination, but configuration was restricted because the current account required an upgrade. No portal permissions were configured in this experiment.

Role-Based Access and Data Sharing Flow

The security flow is: User → User Permissions → Role → Data Sharing Rule → Access Level → Gym Membership Data. It represents how access is controlled in principle and is not evidence that each step was configured.

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO or Gym Manager, which determines the position in the hierarchy. Gym Manager is a conceptual role here; it was not saved in the application.
- A data sharing rule specifies the records shared from one role or user to another.
- The access level (Read / Write) defines what the recipient can do, and this controls access to the Gym Membership data.

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Governance Observations

- Predefined Read and Write permissions were available in the application.
- Roles could be accessed, and a custom role configuration could be prepared.
- Saving the custom role required an upgrade.
- Data Sharing was successfully configured with Read/Write access.
- Portal User Permissions required an upgrade.
- The practical demonstrated how plan restrictions can affect advanced security administration in a low-code platform.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the predefined Read and Write permissions were examined, the role hierarchy with the default CEO role was viewed, and a data sharing rule was successfully configured to share the records of the Gym Membership form from the CEO role to the user Athithya R with Read/Write access.

Some advanced governance capabilities depend on the subscription plan. With the trial expired, saving the Gym Manager custom role and configuring Portal User Permissions were restricted by an upgrade requirement. The experiment therefore shows both the access-control tools that Zoho Creator offers and the effect that plan limits can have on security administration. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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